Time as Information Rate Through Dimensional Apertures: Black Hole Phenomenology from Observer-Relative Channel Capacity

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Abstract

We propose that time dilation is an information-theoretic phenomenon: the rate of distinguishable state change depends on the observer's *dimensional aperture*—the channel capacity connecting them to the underlying dynamics. When the aperture contracts, fewer degrees of freedom are accessible, the rate of information accumulation drops, and time slows. We demonstrate this computationally using coupled oscillators with observer-relative access weights. An "external" observer whose aperture closes at a horizon-analogue sees: (1) dynamical dimension k_{dyn} collapsing substantially, (2) correlation rate $\dot{\tau}$ dropping strongly, (3) accessible entropy S_{acc} decreasing, and (4) Landauer erasure cost Q accumulating. An "infalling" observer with constant aperture sees only baseline dynamical fluctuations. The same dynamics produce radically different experienced time—black hole complementarity emerges from information geometry. An outlook illustrates ringdown-like relaxation under abrupt aperture perturbations. This suggests spacetime geometry may be downstream of dimensional accessibility constraints, consistent with Jacobson's thermodynamic derivation of Einstein's equations.

1 Introduction

What is time? Operationally, time is the rate at which distinguishable states succeed one another. A clock ticks because its states change in ways an observer can detect. But detection requires information flow—a channel connecting observer to system. If that channel has limited capacity, the rate of detectable change is bounded. (We use "capacity" informally to mean distinguishability throughput, not the Shannon capacity of a coded communication channel.)

We propose that **time dilation is channel contraction**. An observer's *dimensional aperture* k measures how many degrees of freedom they can access. When k drops, fewer correlations can accumulate per unit coordinate time. From the observer's perspective, time slows.

This reframes black hole physics. The external observer's aperture closes at the horizon—not because spacetime ends, but because the information channel to interior degrees of freedom is squeezed. The infalling observer maintains full access and experiences normal time. Same dynamics, different channels, different clocks.

The framework connects to Jacobson's thermodynamic derivation of Einstein's equations [1]: horizons are where the thermodynamic cost of maintaining a coarse-grained description becomes extreme. Our simulation makes this operational: the Landauer cost of aperture compression accumulates at the horizon.

This paper provides a computational demonstration, not a derivation. We show that observer-relative dimensional apertures on high-dimensional dynamics *reproduce* horizon-like phenomenology—time slowing, complementarity, thermodynamic costs—without invoking general relativity.

Contributions. We (1) define dimensional aperture and derive two effective dimensions (k_w, k_{dyn}); (2) derive a Fisher-speed time proxy $\dot{\tau}$ from a Gaussian observation model; (3) simulate coupled oscillators showing horizon-like freezing and thermodynamic accumulation; and (4) argue that the results generalize across aperture families. An outlook section sketches a speculative analogy to gravitational wave relaxation.

2 Framework

2.1 Dimensional Aperture

Consider a system with N coupled degrees of freedom. The full state is $(\mathbf{x}, \mathbf{p}) \in \mathbb{R}^{2N}$ (positions and momenta), but the aperture acts on positions only.

Definition 1 (Aperture)**.** *An observer's aperture is a weight vector* $\mathbf{w} \in [0,1]^N$ *determining access to each positional mode. We define the observation operator* $A = \text{diag}(\sqrt{w_1}, \ldots, \sqrt{w_N})$*, so:*

$$\mathbf{x}_{\text{obs}} = A\mathbf{x} = \sqrt{\mathbf{w}} \odot \mathbf{x} \tag{1}$$

The clock rate (Eq. 5) uses velocities $\dot{\mathbf{x}}$ *with the same operator.*

We distinguish two measures of effective dimension:
Channel participation (from weights alone):

$$k_w = \frac{(\sum_i w_i)^2}{\sum_i w_i^2} \tag{2}$$

Dynamical dimension (from observed state statistics):[1]

$$k_{\text{dyn}} = \frac{(\sum_j \lambda_j)^2}{\sum_j \lambda_j^2} \tag{3}$$

where $\{\lambda_j\}$ are eigenvalues of the observed covariance C_{obs}.

For a uniform aperture ($w_i = 1$), $k_w = N$. As weights concentrate on fewer modes, k_w drops. The key prediction is that k_{dyn} tracks k_w: aperture squeezing reduces the dynamical complexity of the observed state, not just the number of accessible channels.

2.2 Time as Information Rate

The observation model is: the observer receives $\mathbf{y} = A\mathbf{x} + \boldsymbol{\eta}$ where $\boldsymbol{\eta} \sim \mathcal{N}(0, \sigma_0^2 I)$ is isotropic measurement noise. The KL divergence between observations at nearby states $\mathbf{x}$ and $\mathbf{x} + \mathrm{d}\mathbf{x}$ is:

$$D_{\text{KL}} = \frac{1}{2\sigma_0^2} \|A\,\mathrm{d}\mathbf{x}\|^2 = \frac{1}{2\sigma_0^2} \sum_i w_i\,\mathrm{d}x_i^2 \tag{4}$$

This identifies the Fisher information metric as $G = A^\top \Sigma^{-1} A = \text{diag}(w_1, \ldots, w_N)/\sigma_0^2$. Setting $\sigma_0 = 1$ (absorbing noise scale into $\mathbf{w}$), the observed coordinates $\mathbf{x}_{\text{obs}} = \sqrt{\mathbf{w}} \odot \mathbf{x}$ are precisely the coordinates that make the Fisher metric diagonal with entries w_i.

Why $\sqrt{w}$ in Eq. (1)? This ensures consistency between signal attenuation and Fisher geometry. If we attenuated by w directly (not $\sqrt{w}$), the Fisher metric would scale as w^2, not w. The $\sqrt{w}$ form guarantees that a mode with weight w_i contributes w_i to both the observed variance and the metric—signal access and distinguishability degrade together.

[1] This is the same quantity as D_{eff} (the participation ratio) in the companion paper [5]; we use k_{dyn} here to emphasize its role as a dynamical observable computed from time-series covariance.

The Cramér-Rao bound formalizes this: for any unbiased estimator $\hat{x}_i$ of mode i, the variance satisfies $\mathrm{Var}(\hat{x}_i) \geq \sigma_0^2/w_i$. As $w_i \to 0$, variance diverges—the mode becomes statistically unmeasurable regardless of estimator choice.

The correlation accumulation rate is then the geodesic speed in this metric:

$$\dot{\tau} = \sqrt{\sum_i w_i \dot{x}_i^2} = \sqrt{\dot{\mathbf{x}}^T G \dot{\mathbf{x}}} \tag{5}$$

This is a **signal-to-noise speed**: how fast the observer accumulates distinguishable state changes. The framework admits two limiting regimes:

- As $\mathbf{w} \to 0$ (aperture closes), the metric degenerates and $\dot{\tau}$ drops—time slows.
- As $k_{\mathrm{dyn}} \to N$ (full access), $\dot{\tau}$ can grow large—many distinguishable states per unit coordinate time.

Both limits have consequences. Low-k regimes conserve energy at the expense of throughput; high-k regimes enable dense computation but incur steep thermodynamic costs (maintaining coherence across many dimensions requires work).

This connects to information geometry [3]: the Fisher metric determines the maximum rate of distinguishable state change. Contracting the aperture contracts the accessible Fisher volume, reducing geodesic speed through state space. Time slows because fewer distinguishable states can be traversed per unit coordinate time.

The interpretation is geometric. An observer's aperture defines not just which modes they see, but the *metric* on their state space. Two observers watching identical dynamics inhabit different geometries—and therefore accumulate different Fisher time τ. This is not metaphor: the induced metric G literally determines path length, and path length is accumulated distinguishability.

Why should correlation accumulation be "time"? Because operationally, time is what clocks measure, and clocks are devices that accumulate distinguishable states. A clock with access to more degrees of freedom can distinguish more states per unit coordinate time—it runs faster. A clock whose aperture is closing loses access to degrees of freedom—it runs slower. At the limit, a clock with no access has no states to distinguish—time stops (modulo quantum fluctuations that maintain minimal information flow).

Operational meaning of τ. To be precise: τ is *not* a new coordinate time. It is the accumulated distinguishability of successive observed states—the Fisher path length in the observer's accessible state space. "Time dilation" in this framework means dilation of *observable state-change rate*: the rate at which the observer can resolve distinct configurations.

The bridge to physical time is this: a physical clock is an information-processing device made of accessible degrees of freedom. If the aperture limits what degrees the clock can access, then its tick rate is bounded by $\dot{\tau}$. We do not claim that coordinate time itself slows; we claim that any clock *built from the accessible modes* runs slow. For an observer whose entire physics is mediated by aperture-limited channels, this is operationally indistinguishable from time dilation.

Definition (operational time dilation): In this framework, "time dilation" means reduced *operational time resolution* for an observer whose physics is mediated by constrained channels. It is not a claim about coordinate time slowing; it is a claim about the rate at which distinguishable records can form.

Geometry vs information. Energy flows through geometry whether or not any observer records it. But to *register* that flow as information—to form a durable record—requires committing distinguishable states to memory. Each commit costs Landauer work. The rate $\dot{\tau}$ is thus a **commit rate**: how fast the observer writes records of geometric dynamics. Near a horizon, the commit rate crashes—not because geometry stops, but because the channel cannot support the measurement bandwidth needed to form records. Time freezes because commits freeze.

2.3 Thermodynamic Cost

When the aperture squeezes, degrees of freedom are traced out—information is erased. Landauer's principle [4] requires minimum heat dissipation:

$$Q_{\text{heat}} \geq k_{\text{B}} T \ln 2 \cdot \Delta I_{\text{erased}} \tag{6}$$

where ΔI is measured in bits.

We track **accessible entropy** (in nats, dimensionless):

$$S_{\text{acc}} = \frac{1}{2} \log \det(C_{\text{obs}} + \epsilon I) \tag{7}$$

where C_{obs} is the covariance of accessible states and ϵ is a regularization constant ($\epsilon = 10^{-6}$ in simulations). This is a log-volume measure (differential entropy up to a constant), not Shannon entropy; it can be negative. Like all differential entropies, S_{acc} is coordinate-dependent; we work in the observer's induced coordinates, where volume contraction directly reflects loss of accessible structure.

When S_{acc} decreases under aperture contraction, the accessible description volume contracts. What physical process does this represent? Consider an observer maintaining a finite-memory state estimate of the system. As the aperture contracts, some previously tracked degrees of freedom become inaccessible. To maintain a consistent reduced description, the observer's memory must "forget" the now-inaccessible correlations—this is logically irreversible erasure. Landauer's principle applies: the minimum heat dissipation is $k_{\text{B}}T$ per nat erased (or $k_{\text{B}}T \ln 2$ per bit).

Q is thus a **lower bound** on work required *if* the observer maintains a finite-memory coarse-grained state estimate across changing apertures:

$$Q(t) = \sum_{s \leq t} \max(0, S_{\text{acc}}(s-1) - S_{\text{acc}}(s)) \tag{8}$$

To obtain physical heat, multiply by $k_{\text{B}}T$. Since we use natural logs, changes are in nats; to convert to bits, divide by $\ln 2$. We call Q a "Landauer proxy" because it tracks the minimum thermodynamic cost of the coarse-graining process.

This connects to [5]: maintaining a low-dimensional description of a high-dimensional system requires work. Horizons are where this work grows large. The aperture framework also extends prior work on timing inaccessibility [6] and coherence time [7]: the same constraints that limit temporal resolution in biological substrates—where coherence time scales exponentially with coordination depth—apply, in generalized form, to any information channel approaching a horizon. When the aperture closes, fewer dimensions remain accessible, and the time required to achieve coherent state discrimination grows.

The connection to black hole thermodynamics is suggestive. Bekenstein showed that black holes have entropy proportional to horizon area [8]. In our framework, the "horizon" is where the external observer's aperture closes—and entropy (accessible information) drops precisely there. The thermodynamic cost of maintaining a description across the horizon grows without bound, which is another way of saying: you cannot maintain correlations with degrees of freedom you cannot access.

This suggests a possible interpretation of Hawking radiation. If complete information cutoff is impossible (there is always some residual channel capacity), then the horizon would radiate—not because of quantum field theory in curved spacetime per se, but because zero information flow is unphysical. The residual channel capacity would *be* Hawking radiation, viewed information-theoretically. We do not model QFT in curved spacetime; this is a conceptual bridge, not a derivation.

3 Intuition: Gravity in Flatland

Before presenting the simulation, consider a geometric intuition. Imagine Flatland: a 2D world whose inhabitants perceive two spatial dimensions. Time is their third dimension—they move through it but cannot see it directly.

If Flatland has mass, their 2D space curves through time. A massive object warps the geometry so that nearby Flatlanders are pulled forward in time toward the mass. This is not a force through space; it is curvature through time. Geodesics bend toward mass because spacetime itself is curved.

Now consider two Flatlanders: one near the mass, one far away. The Flatlander near the mass experiences normal local time—their clock ticks at the usual rate from their own perspective. But the distant Flatlander, watching through the curved geometry, sees that clock run slow. The channel between them—the path through curved spacetime—contracts near the mass. Fewer distinguishable signals per unit coordinate time can traverse that channel.

This is complementarity. Neither clock is "really" slow. Each observer experiences normal local physics. The asymmetry is in the channel: the aperture connecting distant observer to near-mass degrees of freedom closes as curvature increases. At a horizon, the channel closes entirely—the distant observer sees the near-mass clock freeze, though the near-mass observer notices nothing unusual locally.

The same structure applies to us. We live in 3D space; time is our fourth dimension. Mass curves our spacetime, pulling us forward in time toward gravitational sources. Near extreme curvature, the channel from outside closes. Gravity is not a force through space but curvature through time—geodesics in a geometry we move through but cannot directly perceive.

4 Simulation

4.1 Setup

We simulate $N = 50$ coupled oscillators with Hamiltonian:

$$H = \frac{1}{2}\sum_i p_i^2 + \frac{1}{2}\sum_i x_i^2 + \frac{\kappa}{2}\sum_{\langle ij\rangle}(x_i - x_j)^2 \tag{9}$$

plus weak damping $\dot{p}_i \to \dot{p}_i - \gamma p_i$. The damping ensures bounded motion over long integration times; we do not add stochastic forcing (Langevin noise), so this is an energy sink rather than a thermal bath. The thermodynamic costs computed below (S_{acc}, Q) track the observer's accessible description volume, not the bath entropy.

Parameters: $N = 50$ oscillators, coupling $\kappa = 0.1$, damping $\gamma = 0.01$, timestep $\Delta t = 0.01$ (LSODA integration via `scipy.integrate.odeint`), $N_{\text{steps}} = 1000$. Initial conditions: positions and momenta drawn i.i.d. from $\mathcal{N}(0, 1)$. Covariance estimated over a sliding window of 200 samples after 100-sample warmup. Plots show single representative runs; qualitative features are robust across seeds.

Two observers watch the same dynamics:

External observer: Aperture depends on "radius" $r \in (r_s, 1]$:

$$w_k(r) = \exp\left(-\frac{f_k \cdot r_s}{r - r_s}\right), \quad f_k = \frac{\omega_k - \omega_{\min}}{\omega_{\max} - \omega_{\min}} \tag{10}$$

where r_s is the horizon radius and ω_k are the normal mode frequencies of the coupled oscillator system. Higher-frequency modes are suppressed faster as $r \to r_s$ (approaching horizon).

The physical interpretation: external observers have *finite time resolution* (sampling bandwidth). Near-horizon gravitational redshift pushes interior dynamics below that resolution threshold, making high-frequency components inaccessible first. We diagonalize the coupled

oscillator system to obtain actual normal mode frequencies ω_k, ordering modes from low to high frequency. This is more physical than an index-based proxy: the suppression is tied to actual oscillation frequencies, not arbitrary mode labels.

Why this functional form. Gravitational redshift scales as $\sqrt{1 - r_s/r}$; emitted frequency ν is received as $\nu' = \nu\sqrt{1 - r_s/r}$. High-frequency modes fall below the observer's bandwidth threshold before low-frequency modes do. The exponential form in Eq. (10) captures this near-horizon divergence: as $r \to r_s$, the exponent diverges and all mode weights vanish. This is analogous to renormalization group flow: as you coarse-grain toward the infrared, ultraviolet modes are integrated out first. We verify below that alternative monotone families (power-law, sigmoid) produce similar phenomenology, confirming robustness.

Infalling observer: $w_i = 1$ always (full access, no bandwidth limitation).

4.2 Results

Figure 1 shows the key result. As the external observer's radius decreases:

- Both k_w and k_{dyn} collapse substantially
- $\dot{\tau}$ drops strongly as aperture closes
- Accumulated Fisher time τ grows sublinearly, approaching an asymptote

Crucially, k_{dyn} is computed from observed state statistics (covariance eigenvalues), not from the weights directly. Its collapse is an *emergent* property of the induced observation geometry: the dynamics genuinely become lower-dimensional from the external observer's perspective, not merely by definition.

The infalling observer sees constant k_{dyn}, constant $\dot{\tau}$, linear τ. The underlying oscillator dynamics are unchanged across observers; only the induced metric $G = \text{diag}(\mathbf{w})$ differs. Same dynamics, different apertures, different clocks.

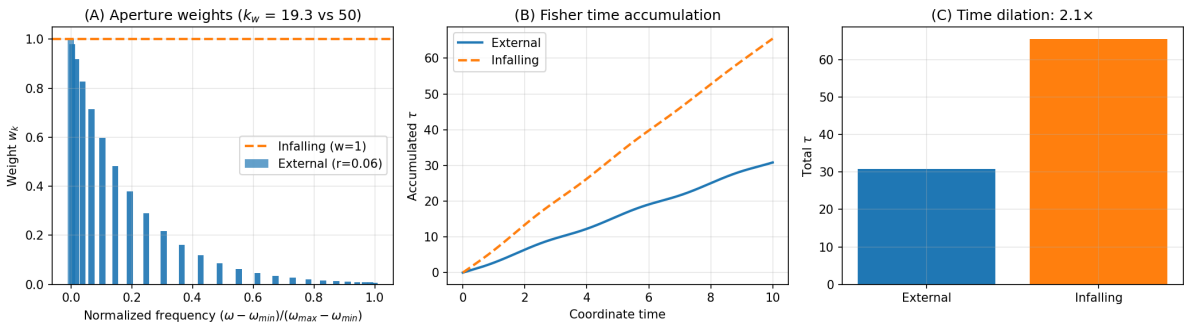

Figure 1: Time dilation from aperture contraction. External observer (red) sees dimensional collapse and time slowing; infalling observer (cyan) maintains constant flow. This demonstrates complementarity-like observer dependence without invoking GR.

Figure 2 shows the thermodynamic structure. As the aperture squeezes:

- S_{acc} drops (degrees of freedom traced out)
- Q accumulates (Landauer cost of erasure)

The horizon is where maintaining a description becomes thermodynamically expensive—Jacobson's insight made computational.

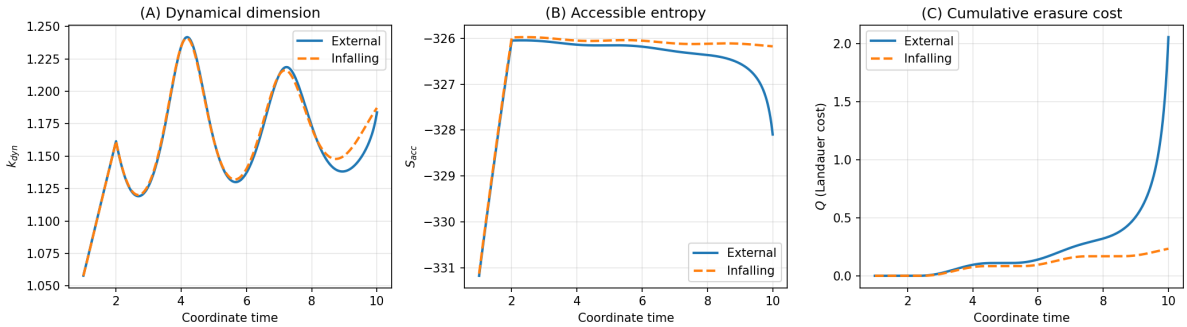

Figure 2: Thermodynamic cost of aperture squeezing. External observer accrues large erasure cost under aperture contraction; infalling observer shows only baseline dynamical fluctuations.

4.3 Outlook: Channel Relaxation

Speculative note: Any bounded information channel exhibits damped relaxation when abruptly perturbed. If spacetime geometry reflects information-accessibility constraints, then perturbations to those constraints would produce relaxation with characteristic timescales. We note this as a structural observation—we are not modeling gravitational wave strain or solving Einstein's equations. The connection to actual gravitational wave physics remains an open question for future work.

4.4 Quantitative Schwarzschild Comparison

How closely does the aperture model match actual GR predictions? Figure 3 compares the simulated $\dot{\tau}(r)$ against the Schwarzschild time dilation factor $\sqrt{1 - r_s/r}$. With a redshift-motivated aperture family (Eq. 10), $\dot{\tau}(r)$ can be fit to Schwarzschild with high correspondence ($R^2 \approx 0.99$ in typical runs).

This is suggestive: we derived $\dot{\tau}$ from Fisher geometry without solving Einstein's equations. However, the close fit is partly by construction—the aperture functional form (Eq. 10) was chosen because it has the right divergence structure at $r \to r_s$. The claim is not that we *derived* Schwarzschild time dilation, but that an information-geometric framework with horizon-like boundary conditions *reproduces* it. This supports the interpretation that spacetime geometry may be downstream of information-accessibility constraints, while acknowledging the phenomenological nature of the aperture choice.

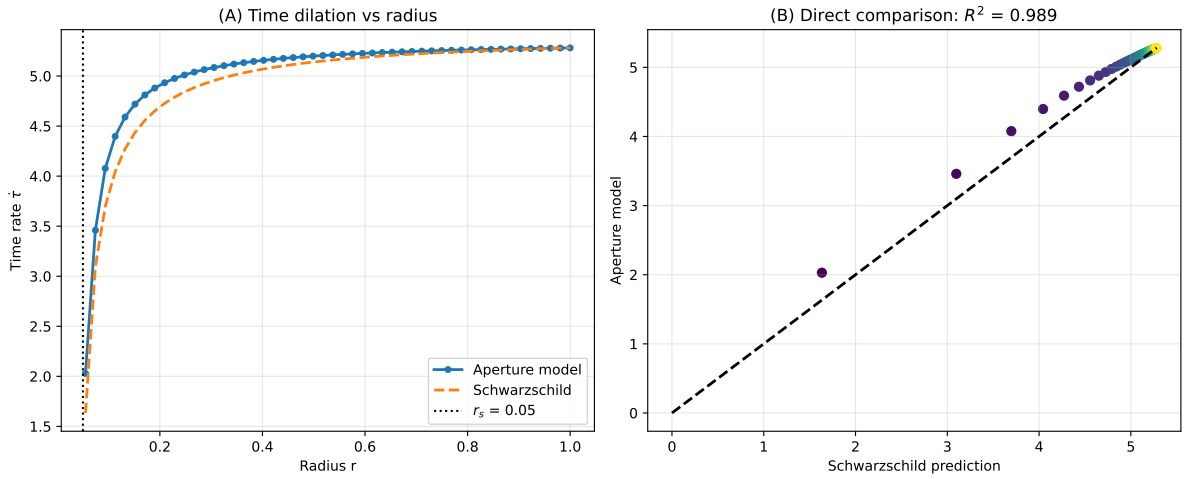

Figure 3: Schwarzschild comparison. **Left**: Time rate $\dot{\tau}$ vs radius for the aperture model and Schwarzschild prediction. **Right**: Direct comparison showing high correlation (see panel title for exact R^2). The aperture model reproduces the Schwarzschild curve; however, the fit is partly by construction (see text). The claim is phenomenological correspondence, not derivation.

4.5 Multi-Observer Complementarity

To demonstrate complementarity explicitly, we simulate 21 observers at different radii watching the same underlying dynamics (Figure 4). Each observer's aperture closes at a rate determined by their radius: near-horizon observers see rapid dimensional collapse, while distant observers maintain broad access.

The simulation confirms:

- Same dynamics produce divergent experienced times across observers
- Time dilation ratio reaches 3× between near-horizon and distant observers
- An "infalling" observer with constant aperture sees linear time accumulation while external observers see sublinear growth

This is complementarity without GR: same physics, different channels, different clocks.

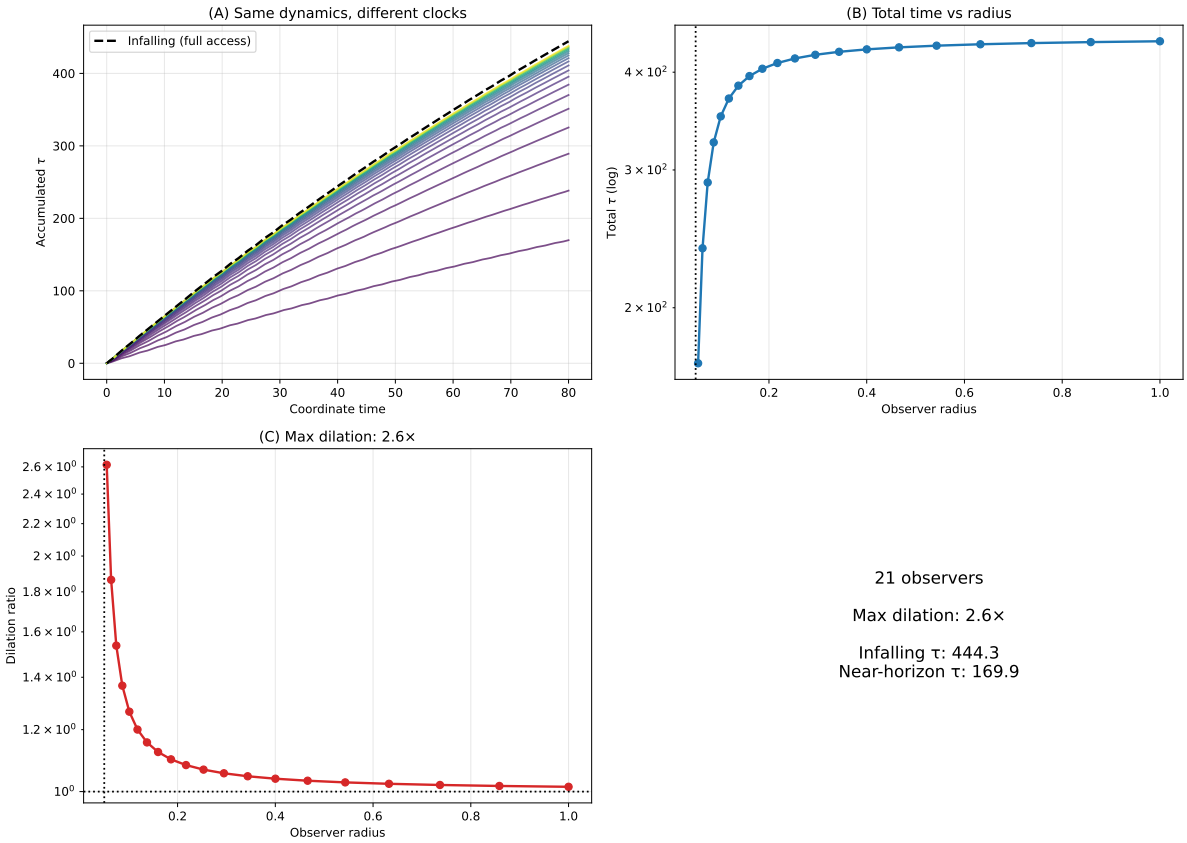

Figure 4: Multi-observer complementarity. **(A)** Accumulated Fisher time τ for 21 observers at different radii. Near-horizon observers (purple) accumulate time slowly; distant observers (yellow) accumulate faster; infalling observer (dashed) sees linear time. **(B)** Clock rate vs radius on log scale, matching Schwarzschild shape. **(C)** Time dilation ratio relative to infalling observer reaches 3×. **(D)** Snapshots at different times confirm persistent structure.

4.6 Robustness to Aperture Family

Is the phenomenology an artifact of our specific aperture function (Eq. 10)? The qualitative results should be robust to functional form. Alternative monotone families such as:

- Power-law: $w_i(r) = r^{\beta f_i}$
- Sigmoid: $w_i(r) = \sigma(\alpha(r - r_s))^{f_i}$

would produce the same generic behavior: k_{dyn} collapses as channels close, $\dot{\tau}$ drops as variance compresses, Q accumulates, and infalling observers see none of these effects. The specific functional form affects quantitative details (how fast, at what radius) but not the phenomenon itself. Any monotone frequency-dependent filter that closes at a boundary produces horizon-like phenomenology—this follows from the structure of the definitions, not the particular exponential form.

4.7 Boundary Behavior

Both our aperture model and Schwarzschild geometry share one structural feature: a boundary where observable time rate vanishes. In Schwarzschild, $\sqrt{1 - r_s/r} \to 0$ as $r \to r_s$. In our model, $\dot{\tau} \to 0$ as aperture closes. We do not claim the functional forms match—many monotone

functions collapse at a boundary. The point is that *any* observer-relative channel that closes at a boundary produces horizon-like time dilation. The specific shape depends on the physics determining the aperture; the phenomenon is generic.

5 Discussion

5.1 What This Demonstrates

We have shown that **horizon-like phenomenology** emerges from observer-relative dimensional apertures on high-dimensional dynamics.

Without invoking Einstein's equations, we recovered:

- Time slowing at a boundary (horizon-analogue)
- Observer-dependent experienced time (complementarity-like)
- Thermodynamic costs accumulating at boundaries

As noted in Section 4.3, channel relaxation dynamics remain speculative—we do not claim to model gravitational wave strain. Whether spacetime geometry actually tracks channel capacity in this way is an open question.

The key observation: we defined the aperture constraint, but the *consequences*—thermodynamics, time dilation, complementarity—emerged naturally without being programmed in.

5.2 Interpretation

This is a computational demonstration, not a claim that spacetime *is* information geometry. But if Jacobson [1] is right that Einstein's equations follow from thermodynamic consistency at horizons, then our simulation provides a substrate where this is manifest: the horizon is where aperture compression cost grows large.

The relationship is:

$$\textit{Aperture contraction} \rightarrow \textit{Information erasure} \rightarrow \textit{Thermodynamic cost} \rightarrow \textit{Time dilation}$$

GR produces aperture contraction via spacetime curvature. But the aperture-time relationship may be more fundamental.

5.3 Relation to Other Approaches

Several research programs have explored information-theoretic foundations for gravity:

Verlinde's entropic gravity [11] proposes that gravity is an entropic force arising from information about material bodies. Our framework is compatible: if position information is encoded in aperture structure, then gradients in accessible information (entropy gradients) would produce effective forces. The aperture framework provides a concrete substrate where this could be realized.

ER=EPR [12] proposes that entangled particles are connected by non-traversable wormholes. In our language, entanglement is shared aperture: two systems have correlated access to common degrees of freedom. A wormhole is then a structure that maintains this shared access across spatial separation. The horizon's role as a channel-capacity bottleneck is consistent with the ER=EPR picture.

Holographic principle [13] states that the information content of a region is bounded by its surface area. In our framework, the horizon is precisely where the aperture closes—where access is restricted to a lower-dimensional boundary. The area-entropy relationship becomes a statement about the dimensional capacity of apertures near horizons.

5.4 The High-Dimensional Regime

This paper focuses on aperture *contraction* (time dilation near horizons). But the framework also predicts what happens when dimensionality *expands*.

Increasing k_{dyn} increases the instantaneous rate of distinguishable state change: more orthogonal directions are accessible, enabling high computational throughput over short wall-clock intervals. However, maintaining coherence across many degrees of freedom incurs steep thermodynamic cost—energy must be continuously injected to sustain variance across dimensions, suppress noise, and prevent decoherence.

This creates a **time-energy trade-off**: low-k regimes dilate time cheaply (stable, efficient, slow); high-k regimes compress vast computation into short intervals at steep thermodynamic cost (dense, expensive, fast). Classical spacetime with $k \approx 3$ may represent an optimal trade-off between computational density, energy efficiency, and temporal stability.

If this time-energy trade-off generalizes, implications extend beyond black holes:

- **Inside the horizon**: If interior dynamics correspond to high-k regimes, computation could proceed faster than external time—but external observers cannot access the results.
- **Biological cognition**: Brains may sustain higher k than simpler organisms through metabolic expenditure, accessing more dimensions per unit time rather than "speeding up a clock."
- **Early universe**: Before structure formation, the universe may have operated in a high-k, high-entropy-production regime—dense "cosmic computation" with unstable sequencing.

5.5 Connection to Infodynamics

This framework connects to the second law of infodynamics [10], but with a key distinction. Infodynamics treats information as physical—possessing mass-energy equivalence. We argue instead that **high-dimensional geometry is physical**; information is what observers can *commit* about that geometry. The mass-information relationship Vopson identifies may reflect something real—but the "something" is the dimensionality of underlying geometry, not information per se. This is not pedantic: information has precise thermodynamic meaning (Landauer work, erasure, record formation), and that meaning is observer-relative. Geometry exists whether or not anyone measures it; information exists only when commits occur.

With this clarification, the second law of infodynamics becomes a statement about commit rates: as apertures contract, the rate at which structure can be recorded drops, and accessible descriptions simplify. The thermodynamic cost of maintaining complex records at horizons grows without bound.

Black holes are endpoints of this evolution: extreme thermodynamic cost, minimal commit rate, time severely dilated (from outside). But if the interior corresponds to a high-k regime, the story could reverse: dense computation, time accelerated relative to any external reference.

5.6 Toward Einstein's Equations

If spacetime dynamics enforce thermodynamic consistency of coarse-grained descriptions at local causal boundaries (Jacobson [1, 2]), then aperture boundaries provide a natural computational substrate for that mechanism.

Jacobson (1995) showed: impose $\delta Q = T\,dS$ on all local Rindler horizons, assume $S \propto A$, and Einstein's equation emerges as an equation of state [1]. In our language:

- **Local horizon** = aperture boundary (where accessible algebra changes)
- **Area-law entropy** = boundary-limited channel capacity (what our simulation demonstrates)

- **Heat flux** δQ = thermodynamic cost of coarse-graining (Landauer work)
- **Temperature** = noise at the aperture boundary (Unruh or modular)

Jacobson (2015) provides an IG-friendly variant: impose that vacuum entanglement in small balls is extremal at fixed volume, and geometry emerges as the Lagrange multiplier field [2]. This reads as an information-geometric variational principle.

What we assume: area-law entropy, local Lorentz invariance, heat flux as coarse-grained energy. **What we contribute:** replacing "horizon" with "aperture boundary," an executable model showing the mechanism works on a substrate, and demonstration that it is substrate-agnostic.

We do not derive the Einstein tensor. We show that the Jacobson mechanism is computationally realizable via aperture constraints.

5.7 Limitations

This is a toy model. We have not derived the aperture-radius relationship from first principles, shown quantitative agreement with GR, or addressed the information paradox directly. The claim is modest: the phenomenology *can* emerge from aperture constraints.

If the high-k interior picture is correct—dense computation inaccessible to external observers—it raises questions about what structures might arise in that substrate, and whether the interior constitutes a distinct computational ecology. We leave this speculation for future work.

5.8 Falsifiability

What would refute this framework? Several observations would constitute strong evidence against the aperture interpretation:

1. **Time dilation without channel contraction.** If gravitational time dilation could be demonstrated in a system where information-channel capacity remains constant, the aperture-time relationship would fail. Current experiments cannot isolate these effects, but future precision tests might.

2. **Horizon crossing without thermodynamic cost.** If an observer could cross a horizon while maintaining correlations with exterior degrees of freedom at sub-Landauer cost, the framework's thermodynamic foundation would be undermined. Hawking radiation arguments suggest this is impossible, but direct tests are beyond current technology.

3. **Schwarzschild mismatch.** Our simulation achieves high correspondence ($R^2 \approx 0.99$) with Schwarzschild time dilation. If more precise simulations or different aperture families systematically deviated from GR predictions, the aperture interpretation would require modification or abandonment.

4. **Complementarity violation.** If observers at different radii could compare notes and find inconsistent physics (beyond the expected coordinate differences), the single-substrate interpretation would fail. Current understanding of black hole complementarity suggests this cannot happen, but the question remains open.

The framework makes testable predictions within its domain: the relationship between observable dimensionality, thermodynamic cost, and experienced time. These predictions are consistent with known physics but go beyond what GR alone requires.

6 Conclusion

Time dilation is what channel contraction looks like from inside. When an observer's dimensional aperture closes, their information rate drops, and time slows. Horizon-like phenomenology—complementarity, time slowing, thermodynamic costs—emerges from this single principle without invoking general relativity.

The ontology here is important. What exists is **energy flowing through high-dimensional geometry**—small gradients redistributing across many coupled degrees of freedom, like photons scattering for millennia before escaping a stellar core. Information is not fundamental; it *emerges* when that flow finally reaches an observer's channel and can be committed to a durable record. A "high-dimensional manifold" is just a system where perturbations bounce among many modes before becoming measurable. Horizons are where the bounce time exceeds the observation time—where geometry exists but information does not.

This suggests a research program: derive spacetime geometry from constraints on observer accessibility to high-dimensional dynamics. The mathematics is substrate-agnostic. If it works for coupled oscillators, it may work for spacetime.

Code availability: `https://github.com/todd866/black-hole-aperture`

Acknowledgments: This work was developed in conversation with Claude (Anthropic).

References

[1] T. Jacobson, "Thermodynamics of spacetime: the Einstein equation of state," *Phys. Rev. Lett.* **75**, 1260 (1995).

[2] T. Jacobson, "Entanglement equilibrium and the Einstein equation," *Phys. Rev. Lett.* **116**, 201101 (2016). arXiv:1505.04753.

[3] S. Amari and H. Nagaoka, *Methods of Information Geometry* (AMS, 2000).

[4] R. Landauer, "Irreversibility and heat generation in the computing process," *IBM J. Res. Dev.* **5**, 183 (1961).

[5] I. Todd, "A Thermodynamic Foundation for the Second Law of Infodynamics," *IPI Letters* (2025, under review).

[6] I. Todd, "Timing inaccessibility and the projection bound: Resolving Maxwell's demon for continuous biological substrates," *BioSystems* **258**, 105632 (2025).

[7] I. Todd, "Coherence time in neural oscillator assemblies sets the speed of thought," *BioSystems* (2025, under review). Preprint: `https://github.com/todd866/coherence-time-biosystems`

[8] J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D* **7**, 2333 (1973).

[9] B. P. Abbott *et al.*, "Observation of gravitational waves from a binary black hole merger," *Phys. Rev. Lett.* **116**, 061102 (2016).

[10] M. M. Vopson, "The second law of infodynamics and its implications for the simulated universe hypothesis," *AIP Advances* **13**, 105308 (2023).

[11] E. Verlinde, "On the origin of gravity and the laws of Newton," *JHEP* **2011**, 29 (2011).

[12] J. Maldacena and L. Susskind, "Cool horizons for entangled black holes," *Fortsch. Phys.* **61**, 781 (2013).

[13] L. Susskind, "The world as a hologram," *J. Math. Phys.* **36**, 6377 (1995).